Appendix

A Sources of every number we entered (sources.csv)

sources.csv has 138 entries (29 public, 43 by analogy, 66 guesstimate). The table shows the 51 entries of the files that drive the results: product, suppliers, routing, lanes, freight rates, emission factors. The other 87 entries (bom.csv 27, workcenters.csv 30, process_environment.csv 20, locations.csv 10) are in the submitted sources.csv with the same columns (including the sanity check). “Guesstimate” means our own estimate with the reasoning given.

file	row	column	value	type	source or calculation
products.csv	CHAIR-EVERY-EV15R	mass_kg	**18.5**	public	Interstuhl EVERY POLSTER official web shop spec (Produktgewicht 18.5 kg) (https://shop.interstuhl.com/de-de/produkte/every/aktionsmodell-polster-ev15r?edition=01)
	CHAIR-EVERY-EV15R	list_price_eur	**310.08**	public	Interstuhl EVERY POLSTER official web shop price 369.00 EUR incl 19% VAT (369.00 / 1.19 = 310.08 EUR net) (https://shop.interstuhl.com/de-de/produkte/every/aktionsmodell-polster-ev15r?edition=01)
suppliers.csv	SUP-CN	capacity_per_month	**15000.0**	guesstimate	Assumed monthly supplier capacity allocation; supplier can easily deliver this capacity to cover plant demand
	SUP-DE	capacity_per_month	**12000.0**	guesstimate	Assumed monthly supplier capacity allocation; supplier can easily deliver this capacity to cover plant demand
	SUP-CN	lead_time_mean_days	**35.0**	guesstimate	Calculated in calendar days: 5.0 prep + 0.5 truck (LANE-CN-SHA 165 km) + 3.0 Shanghai port + 35.0 ocean (LANE-SHA-RTM 21989 km) + 3.0 Rotterdam port + 1.5 truck (LANE-RTM-PLANT 680 km) = 48.0 calendar days; 48.0 x 5/7 = 34.3, rounded to 35.0 working days (the model clock counts working days)
	SUP-DE	lead_time_mean_days	**4.5**	guesstimate	Calculated: 3.5d order processing & factory dispatch + 1.0d direct truck (LANE-DE-PLANT 360km) = 4.5 working days (domestic: no conversion needed)
	SUP-CN	promised_lead_days	**30.0**	analogy	Contractual baseline lead time agreed with Asian supplier (30 working days)
	SUP-CN	lead_time_sd_days	**3.5**	guesstimate	Estimated standard deviation of 3.5 days (~10% of 35-day mean lead time)
	SUP-DE	promised_lead_days	**4.0**	analogy	Contractual lead time agreed with domestic supplier Stabilus (4 working days)
	SUP-DE	lead_time_sd_days	**1.0**	guesstimate	Estimated standard deviation of 1.0 day for domestic truck transit
	SUP-CN	kg_co2e_per_unit	**3.26**	guesstimate	Calculated: 1.15 kg steel * 1.92 kg CO2/kg + 2.0 kWh * 0.525 kg CO2/kWh (grid_electricity_CN_2025) = 3.258 ~ 3.26 kg CO2e
	SUP-DE	kg_co2e_per_unit	**2.91**	guesstimate	Calculated: 1.15 kg steel * 1.92 kg CO2/kg + 2.0 kWh * 0.353 kg CO2/kWh (grid_electricity_DE_2024) = 2.914 ~ 2.91 kg CO2e
	SUP-DE	unit_price_eur	**22.0**	guesstimate	Estimated Stabilus B2B volume price based on Ludwig Meister listing (27.41 EUR/unit at 55+ pcs) https://www.ludwigmeister.de/de/sortiment/edi/4919823
	SUP-CN	unit_price_eur	**10.0**	guesstimate	B2B industry benchmark price of 10.00 EUR for Class 4 gas lift cylinder at high volume (supplier list price is price on request)
routing.csv	WC1	process_time_min	**5.0**	guesstimate	Interstuhl video (07:13-08:20): PU foam injection into mould + curing time + manual edge trimming with electric shears (08:05-08:20)
	WC1	setup_time_min	**0.0**	analogy	0 setup time in baseline
	WC2	process_time_min	**4.5**	guesstimate	Interstuhl video (09:29-10:01): Position Manhattan fabric + pneumatic stapling (113 staples) + string tensioning + heat-gun wrinkle smoothing
	WC2	setup_time_min	**0.0**	analogy	0 setup time in baseline
	WC3	process_time_min	**7.2**	guesstimate	Interstuhl video (10:01-10:42 & 18:00): Mount Autolift mechanism (120s) + insert EVERY gas spring into star base & casters (120s) + bolt main frame (192s) = 432s = 7.2 min
	WC3	setup_time_min	**0.0**	analogy	0 setup time in baseline
	WC4	process_time_min	**4.2**	guesstimate	Interstuhl video (12:00-13:15): Mount padded backrest & 2D armrests (150s) + test Autolift height lock & tilt mechanism (102s) = 252s = 4.2 min
	WC4	setup_time_min	**0.0**	analogy	0 setup time in baseline
	WC5	process_time_min	**2.1**	guesstimate	Interstuhl video (13:15-13:45): Fold carton, insert chair parts + EPS caps + manual + tape seal (126s) = 2.1 min
	WC5	setup_time_min	**0.0**	analogy	0 setup time in baseline
network_lanes.csv	LANE-CN-SHA	distance_km	**165.0**	public	Road distance Ningbo to Port of Shanghai maps.openrouteservice https://maps.openrouteservice.org/#/directions/Ningbo,ZJ,Volksrepublik%20China/Shanghai,SH,Volksrepublik%20China/... (full link in sources.csv)
	LANE-CN-SHA	transit_days_mean	**0.5**	guesstimate	165 km / 75 km/h = 2.2 h driving + port handling = 0.5 d
	LANE-CN-SHA	transit_days_sd	**0.2**	analogy	0.2 days variability for short road haul
	LANE-SHA-RTM	distance_km	**21989.0**	public	11873 nautical miles via Suez x 1.852 = 21989 km
	LANE-SHA-RTM	transit_days_mean	**35.0**	public	Shanghai to Rotterdam ocean transit schedule (35 calendar days) https://searoutesnav.com/r/6nS7zAifMven
	LANE-SHA-RTM	transit_days_sd	**3.0**	analogy	3.0 days standard deviation via Suez
	LANE-RTM-PLANT	distance_km	**680.0**	public	Port of Rotterdam to Messstetten plant road distance maps.openrouteservice https://maps.openrouteservice.org/#/directions/Rotterdam,ZH,Niederlande/Kreuzb%C3%BChlstra%C3%9Fe,Me%C3%9Fstetten,BW,Deutschland/... (full link in sources.csv)
	LANE-RTM-PLANT	transit_days_mean	**1.5**	guesstimate	680 km / 75 km/h = 9.0 h driving + rest rules = 1.5 d
	LANE-RTM-PLANT	transit_days_sd	**0.3**	analogy	0.3 days standard deviation for truck transit
	LANE-DE-PLANT	distance_km	**360.0**	public	Koblenz to Messstetten plant road distance maps.openrouteservice https://maps.openrouteservice.org/#/directions/Koblenz,RP,Deutschland/Kreuzb%C3%BChlstra%C3%9Fe,Me%C3%9Fstetten,BW,Deutschland/... (full link in sources.csv)
	LANE-DE-PLANT	transit_days_mean	**1.0**	guesstimate	360 km / 75 km/h = 4.8 h driving + loading = 1.0 d
	LANE-DE-PLANT	transit_days_sd	**0.2**	analogy	0.2 days standard deviation for domestic truck
freight_rates.csv	LANE-CN-SHA	rate_eur_per_shipment	**120.0**	analogy	Fixed drayage truck rate Ningbo to Shanghai
	LANE-CN-SHA	rate_eur_per_tkm	**0.1**	analogy	0.10 EUR/tkm variable truck rate
	LANE-CN-SHA	min_charge_eur	**90.0**	analogy	90 EUR minimum shipment charge
	LANE-SHA-RTM	rate_eur_per_shipment	**3686.0**	public	Drewry Container Index Shanghai-Rotterdam https://www.drewry.co.uk/supply-chain-advisors/supply-chain-expertise/world-container-index-assessed-by-drewry
	LANE-SHA-RTM	rate_eur_per_tkm	**0.18**	guesstimate	0.18 EUR/tkm flat container pricing
	LANE-SHA-RTM	min_charge_eur	**3686.0**	guesstimate	Minimum charge equals container spot rate
	LANE-RTM-PLANT	rate_eur_per_shipment	**120.0**	analogy	120 EUR fixed truck booking fee Rotterdam to Messstetten
	LANE-RTM-PLANT	rate_eur_per_tkm	**0.1**	analogy	0.10 EUR/tkm hinterland road freight rate
	LANE-RTM-PLANT	min_charge_eur	**90.0**	analogy	90 EUR minimum charge
	LANE-DE-PLANT	rate_eur_per_shipment	**150.0**	analogy	150 EUR fixed truck booking fee Koblenz to Messstetten
	LANE-DE-PLANT	rate_eur_per_tkm	**0.09**	analogy	0.09 EUR/tkm direct domestic truck rate
	LANE-DE-PLANT	min_charge_eur	**100.0**	analogy	100 EUR minimum charge
emission_factors.csv	grid_electricity_DE_2024	value	**0.353**	public	Umweltbundesamt (UBA) CO2-Emissionsfaktor Strommix Deutschland 2024 (0.353 kg CO2e/kWh) (https://www.umweltbundesamt.de/themen/klima-energie/energieversorgung/strom-waermeversorgung-in-zahlen)
	crude_steel_global_2024	value	**1.92**	public	worldsteel Sustainability Indicators 2024 report (1.92 kg CO2e per kg crude steel) (https://worldsteel.org/wider-sustainability/sustainability-indicators/)
	grid_electricity_CN_2025	value	**0.525**	public	Our World in Data / Ember Carbon Intensity of Electricity China 2025 (0.525 kg CO2e/kWh) (https://ourworldindata.org/grapher/electricity-mix?mapSelect=%7ECHN&source=total&metric=carbon_intensity&frequency=annual)

B Check cell output (v1.0, final data files, baseline seeds 1–10)

```
PASS  T1 Format
      all 10 files present, columns exact, no blanks or negatives
PASS  T2 Capacity
      WC1 PU Foam Moulding & Trimming: busy 81.0% of available time
      WC2 Upholstery & Fabric Cover Assembly: busy 72.9% of available time
      WC3 Gas Spring Base Casters & Tilt Mechanism Sub-Assembly: busy 77.8% of available time
      WC4 Backrest Joining Armrest & Ergonomic Test: busy 66.6% of available time
      WC5 Packaging: busy 68.1% of available time
PASS  T3 Little's Law
      run 1: L = 12.62  vs  λ×W = 17.497/h × 0.72 h = 12.60  (gap 0.2%)
      10 run(s) checked
PASS  T4 Flow balance
      run 1: started 36,400, finished 36,393, key components used 36,396 (allowed 36,393 to 36,400)
PASS  T5 Margin
      cost per unit 149.93 to 149.94 EUR vs list price 310.08 EUR
------------------------------------------------------------------------------
5 of 5 tests passed  ·  ALL PASS
check cell v1.0 · cell 3ad30383c119 · data b923a5690b12 · 2026-11-10 22:08
```

C The key argument of each paper in three sentences

Ambulkar, Blackhurst & Grawe (2015)

The authors ask how a firm becomes resilient to supply chain disruptions and build a scale to measure firm resilience. They find that a disruption orientation alone is not enough: after a high-impact disruption its effect on resilience runs entirely through reconfiguring resources, after a low-impact one it works together with risk-management infrastructure. This explains our result that the stand-still falls from 133 to 8 hours: we treat the Suez closure as high-impact, and the switch to Stabilus in week 16 is the reconfiguration.

Schoenherr, Mena, Vakil & Choi (2023)

Based on interviews with purchasing and supply professionals, the authors ask how measuring risk can make supply chains more resilient. They find that risk must be measured continuously, that the measurements must lead to action, and that this takes a culture of measuring rather than an occasional check. This explains how we read the weeks of spring cover (2.4, 1.8, 0.1 in weeks 15, 16, 18): it is an early signal that makes the switch in week 16 timely, while the stand-still hours only show the problem in weeks 19 to 21.

Ambulkar, S., Blackhurst, J., & Grawe, S. (2015). Firm's resilience to supply chain disruptions: Scale development and empirical examination. *Journal of Operations Management, 33–34*, 111–122. https://doi.org/10.1016/j.jom.2014.11.002

Schoenherr, T., Mena, C., Vakil, B., & Choi, T. Y. (2023). Creating resilient supply chains through a culture of measuring. *Journal of Purchasing and Supply Management, 29*(4), 100824. https://doi.org/10.1016/j.pursup.2023.100824

D Note on the use of AI

We used Claude (Anthropic) as an assistant. Claude wrote the first version of the simulation notebook (engine, three scenarios, results.json, animation), revised it after the checkpoint feedback, helped to collect and document the sources in sources.csv, and designed and laid out the poster. We checked the output with the check cell (five tests, among them Little's Law and flow balance) and by comparing every number on the poster with the results.json of our notebook. One mistake we found: Claude's first Stabilus price of 14.50 EUR was a "B2B premium quote" without any calculation and made the German supplier look cheaper than our baseline. The checkpoint re-check pointed to it. We replaced it with an estimate from a listing (27.41 EUR, discounted to 22.00 EUR), re-ran all scenarios, and the German supplier now costs 2.61 EUR more per chair. The model assumptions marked "our assumption" on the poster (12-week closure, switch in week 16, Suez passage at 65 % of the lead time), the choice of the real event and the reading of the two papers are the team's decisions.